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Budget-Aware Adaptive LoRA: Gradient-Driven Rank Allocation for Parameter-Efficient Fine-Tuning

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INTRODUCTION

Large language models (GPT, BERT, LLaMA) are costly in terms of computation, as it is required to update billions of parameter to fine tune the full model. Adaptation of these LLMs for a task specific problem is prohibitively expensive for most researchers and organizations. **LoRA** (Low-Rank Adaptation) reduces the trainable parameters using small adapter matrices, and uniform rank across all layers.

- Traditional fine-tuning of LLMs:

  $W_{new} = W_{pretrained} + \Delta W$; (ΔW is huge as $W_{pretrained}$)

- LoRA's approach:

  $W_{new} = W_{pretrained} + B \times A$; (B&A are very small W)

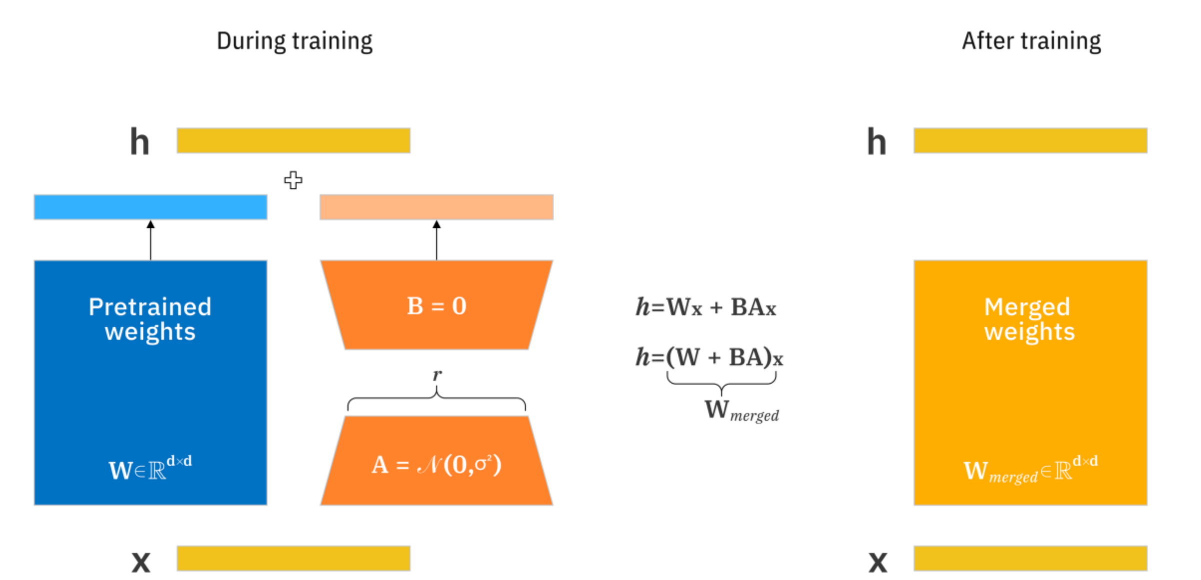

Figure 1: *The workflow of LoRA with adapter matrices*

LoRA represents weight updates in a low intrinsic dimension. In stead of updating the (12,288 × 12,288 = 150M), LoRA play with a smart trick of two tiny matrices B(12,288 × 8) & A(8 × 12,288)which turns the billions of trainable parameters into just 196K. Here, 8 is the rank which is uniform across all the layers.

Research Question

Does adaptive rank allocation improve upon uniform LoRA under strict parameter budgets?

LoRA reduces the trainable parameters and GPU usage in a massive number. However, it lacks on some points:

The rank allocation is not optional, since it assigns rank uniformly. Finding the right rank need grid search over r ε {2,4,8,16,32,64}which is costly.

BA LoRA's adaptive allocation strategy combines

(1) Gradient based layer importance estimation. (2) Budget-aware adaptive rank allocation. (3) Single pass training (comparable efficiency)

Method	Adaptive	Budget	Single Pass
LoRA	❌	✅	✅
AdALoRA	✅	❌	❌
GoRA	✅	❌	✅
BA-LoRA	✅	✅	✅

Table 1: *Features eligibility of LoRA, ALoRA, GoRA, and BA-LoRA*

RESULTS

Experiment: BA-LoRA's gradient-based adaptive allocation VS uniform LoRA across 4datasets (SST2, IMDB, Tweet Eval, AG News) with ranks r ∈ {2,4,8,12} and parameter budgets of 630K-815K.

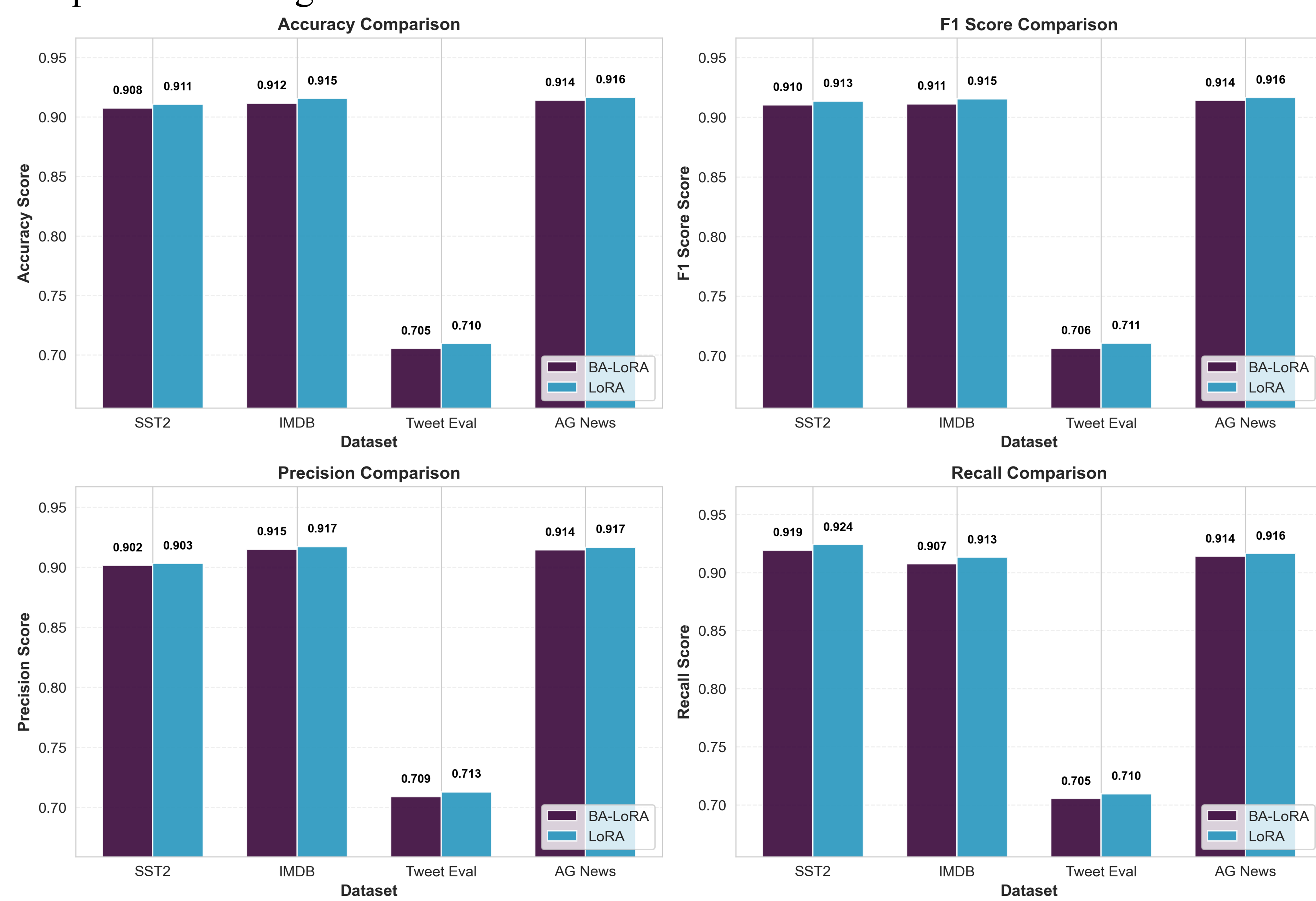

Figure3: BA-LoRA vs LoRA: Multi-Dataset Performance Comparison

Research Question: Does adaptive allocation beat uniform LoRA?
Answer: No (not on text classification with base models)

Adaptive rank allocation does NOT improve upon uniform LoRA for text classification with base models. We assume that layer homogeneity, task simplicity, and tight parameter constraints limit the benefits of adaptive allocation. Simple uniform LoRA is simpler to implement, equally or more effective, and should be the default choice for practitioners.

Dataset	BA-LoRA Acc	LoRA Acc	Δ Acc	p-value	Significant	Time Overhead
SST2	0.9076 ± 0.0255	0.9107 ± 0.0254	-0.0031	0.7634	✗	9.8%
IMDB	0.9116 ± 0.0212	0.9154 ± 0.0200	-0.0038	0.6381	✗	-21.1%
TWEET_EVAL	0.7052 ± 0.0142	0.7095 ± 0.0122	-0.0043	0.4326	✗	-0.7%
AG_NEWS	0.9141 ± 0.0037	0.9165 ± 0.0006	-0.0024	0.3845	✗	-1.2%

Table2: statistical summary

Broader Impact: This rigorous evaluation validates that simple baselines are highly effective. Negative results prevent unnecessary complexity in future PEFT methods and guide research toward domains where adaptivity may actually help (e.g., larger models, complex reasoning tasks).

Future Work:
- Test on larger models (LLaMA-7B/70B) where layer specialization may matter more
- Evaluate on complex tasks: reasoning, code generation, multi-step QA
- Explore advanced importance metrics: parameter sensitivity, activation statistics
- Investigate wider rank allocation ranges beyond [0.5×, 2.0×]

METHODS

We evaluated BA-LoRA against vanilla LoRA across 4 datasets and 2 model architectures in 103 experiments.

Datasets & Models: **SST-2** (67K train), **IMDB** (25K train), **AG News** (120K train), **TweetEval** (45K train) • **DistilBERT-base** (66M params) & **RoBERTa-base** (125M params) • Ranks tested: r ∈ {2, 3, 4, 6, 8, 12}

BA-LoRA Pipeline:

1. ***Gradient Importance:*** $I(W) = \text{avg}(|W \odot \nabla W|)$ on 5K samples
2. ***Budget-Aware Allocation:*** Ranks ∈ [0.5×, 2.0×] base_rank Iterative adjustment → exact budget
3. ***Standard Fine-Tuning:*** Random init, AdamW, lr=5e-4, epochs=3

Evaluation: Accuracy, F1, Precision, Recall | Statistical testing: Welch's t-test (α=0.05)

DISCUSSION

Why Didn't Adaptive Allocation Help? **Several factors may explain the comparable performance:**

- Task Simplicity— All four are classification tasks; uniform allocation may suffice for learning decision boundaries.
- Model Scale — Benefits may emerge at larger scales (7B+ params); tested base models (66M-125M).
- Importance Metric Limitations — Gradient magnitude may not fully capture layer importance; other metrics may work better.
- Strong Baseline — Uniform LoRA surprisingly effective; simple methods often work well with sufficient data.

Despite comparable performance, this work provides valuable insights:

- Fair Comparison Framework— First large-scale study (103 exp) with strict parameter budget enforcement
- Comprehensive Evaluation — 4 datasets, 2 models, diverse tasks (sentiment, topic classification)
- Negative Results Matter — Validates uniform LoRA's robustness; prevents overengineering
- Efficiency Insights — BA-LoRA faster on 3/4 datasets; overhead assumptions don't always hold

References

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